# Hanpeng Liu

Ph.D. Candidate, School of Computer Science and Technology
Huazhong University of Science and Technology (HUST), Wuhan, China

hanpengliu@hust.edu.cn · 1359947339@qq.com
Homepage · Google Scholar · GitHub

## Research Summary

My research focuses on visual representation learning for multimodal large language models (MLLMs), spanning image-text pretraining [C1], CLIP post-training [C2], self-supervised visual pretraining [J1, P1], and instruction-aware vision encoding for multimodal understanding [P2]. My publications include three NeurIPS papers and one CVPR paper; I am first or co-first author on four papers and corresponding author on one. Going forward, I aim to extend this line of work toward multimodal understanding or image generation.

**Interests:** Vision encoders for multimodal LLMs · Vision-language pretraining (CLIP) · Self-supervised visual representation learning · Efficient architectures and knowledge distillation

## Education

**Huazhong University of Science and Technology (HUST)** Sep 2022 – Jun 2027 (expected)
*Ph.D. in Computer Science and Technology (direct-entry program); Advisor: Prof. Kun He* *Wuhan, China*

**Huazhong University of Science and Technology (HUST)** Sep 2018 – Jun 2022
*B.Eng. in Computer Science and Technology; Rank: 23/373; Outstanding Graduate* *Wuhan, China*

## Publications

[*] Equal contribution [†] Corresponding author

### Conference Papers

[C1] **Hanpeng Liu**, Yaqian Li[†], Zidan Wang, Shuoxi Zhang, Zonglin Zhao, Zihao Bo, Rinyoichi Takezoe, Kaiwen Long, Kun He[†]. *ITO: Multi-View Alignment and Training-Time Fusion for Image-Text Pretraining*. Advances in Neural Information Processing Systems (**NeurIPS**), 2026. [Paper] [Code] [Project]

[C2] Zidan Wang, Yaqian Li[†], Xiaokai Zhang, Kaiwen Long, Kun He, **Hanpeng Liu**[†]. *Rethinking Contrastive Loss in CLIP Post-training: A Complementary Framework with Frozen Text Encoder*. Advances in Neural Information Processing Systems (**NeurIPS**), 2026. [Paper] [Code] [Project]

[C3] Shuoxi Zhang, **Hanpeng Liu**, Stephen Lin, Kun He[†]. *Hyperspherical Dataset Distillation via Contrastive Embedding Alignment*. IEEE International Conference on Multimedia and Expo (**ICME**), 2025. [Paper]

[C4] Jinsong Chen, **Hanpeng Liu**, John E. Hopcroft, Kun He. *Leveraging Contrastive Learning for Enhanced Node Representations in Tokenized Graph Transformers*. Advances in Neural Information Processing Systems (**NeurIPS**), 2024. [Paper]

[C5] Shuoxi Zhang, **Hanpeng Liu**, Yuyi Wang, Kun He[†]. *Grouped Logit Distillation Enhanced with Superclass Awareness for Efficient Knowledge Transfer*. European Conference on Artificial Intelligence (**ECAI**), 2024. [Paper]

[C6] Shuoxi Zhang[*], **Hanpeng Liu**[*], Stephen Lin, Kun He. *You Only Need Less Attention at Each Stage in Vision Transformers*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024. [Paper]

### Journal Papers

[J1] **Hanpeng Liu**, Shuoxi Zhang, Kaiyuan Gao, Kun He[†]. *Boosting the Patch-based Self-Supervised Learning through Past-to-Present Smoothing*. **Pattern Recognition**, 2025.

[J2] Shuoxi Zhang, **Hanpeng Liu**, Yuyi Wang, Kun He, Jun Lin, Yang Zeng. *Class Discriminative Knowledge Distillation*. **IEEE Transactions on Emerging Topics in Computational Intelligence**, 9(2):1340–1351, 2025. [Paper]

[J3] **Hanpeng Liu**[*], Shuoxi Zhang[*], Kun He[†]. *Residual Learning for Self-Knowledge Distillation*. **IEEE Transactions on Big Data**, 2025.

[J4] Shuoxi Zhang, **Hanpeng Liu**, Kun He. *Knowledge Distillation via Token-Level Relationship Graph Based on the Big Data Technologies*. **Big Data Research**, 2024.

**Preprints**

[P1] **Hanpeng Liu**, Zidan Wang, Shuoxi Zhang, Kaiyuan Gao, Kun He. *Separators in Enhancing Autoregressive Pretraining for Vision Mamba*. arXiv:2603.03806, 2026. [Paper]

[P2] **Hanpeng Liu**, Yaqian Li†, Zidan Wang, Shuoxi Zhang, Zihao Bo, Rinyoichi Takezoe, Kaiwen Long, Kun He†. *iGVLM: Dynamic Instruction-Guided Vision Encoding for Question-Aware Multimodal Understanding*. arXiv:2603.02748, 2026. [Paper]

## Research Experience

**Li Auto Inc.** Sep 2024 – Present
*Research Intern* *China*

- Helped build a distributed training and evaluation platform for vision and vision-language foundation models (DDP / FSDP / DeepSpeed), with standardized benchmarks on ImageNet and COCO.
- Led the scaling of CLIP-style image-text pretraining to **128 NVIDIA A800 GPUs** on **1B-scale** image-text data, using gradient-preserving all-gather for the contrastive loss and WebDataset streaming pipelines.

## Mentoring

- **CLIP post-training** [C2] (NeurIPS 2026, co-corresponding author): proposed the research idea, mentored junior HUST students through the project, contributed to the experiments, and led the paper writing.

## Academic Service

- **Conference Reviewer:** NeurIPS, ICLR, CVPR, AAAI
- **Journal Reviewer:** Pattern Recognition, IEEE Transactions on Cybernetics

## Honors and Awards

| | |
|---|---|
| Merit Graduate Student & Zhixing Scholarship, HUST | 2023, 2024 |
| Outstanding Graduate, HUST | 2022 |
| Third Prize (National), China Post-Graduate Mathematical Contest in Modeling | 2022 |
| Third Prize (National), Bisheng Cup Compiler System Design Competition (Team Leader) | 2021 |
| First Prize (National), Chinese Mathematics Competitions | 2020 |

## Skills

**Programming:** Python, PyTorch, timm, Hugging Face Transformers

**Large-scale training:** Multi-GPU / multi-node distributed training (DDP, FSDP, DeepSpeed ZeRO), mixed-precision training, WebDataset streaming pipelines; LLaVA-style MLLM training with Qwen2 backbones

## References

**Prof. Kun He**
Professor, School of Computer Science and Technology
Executive Director, Hopcroft Center on Computing Science
Huazhong University of Science and Technology
brooklet60@hust.edu.cn
*Ph.D. advisor*

**Yaqian Li**
R&D Architect, Compute Architecture
Li Auto Inc.
*Industry collaborator; co-corresponding author on [C1], [C2], [P2]*

Additional references available upon request.